

UNF

UNF is the beginning process for database normalization, it is composed of the list of attributes with the indication of repeating data for group of attributes.

Here, the repeating group is the collection of related attributes whose values can be repeated multiple times.

All the attributes with their respective entities are listed out the repeating group are indicated using curly braces.

Employee (client_employee_id, first_name, middle_name, last_name, preferred_name, job_code, job_title, job_start_date, { organization_id, organization_name }, { department_id, department_name }, dob, hire_date, recent_hire_date, anniversary_date, term_date, years_of_experience, work_email, address, city, state, zip, country, { manager_employee_id, manager_employee_name }, fte_status, is_per_deim, cell_phone, work_phone, scheduled_weekly_hour, active_status, termination_reason, clinical_level)

Here the organization, department and manager group are repeating group and are indicated using {} bracket.

1NF

The entity is in first normal form if it doesn't contain any repeating groups and single value or each row properties are held. In first normal form, the repeating group are separated into a new entity, the unique key for each new entity is identified and relationship between the two or more tables are created by addition of the foreign keys.

For Employee Entity

Employee (client_employee_id (PK), first_name, middle_name, last_name, preferred_name, job_code, job_title, job_start_date, dob, hire_date, recent_hire_date, anniversary_date, term_date, years_of_experience, work_email, address, city, state, zip, country, fte_status, is_per_deim, cell_phone, work_phone, scheduled_weekly_hour, active_status, termination_reason, clinical_level, organization_id (FK), department_id (FK), manager_employee_id (FK))

Primary Key: client_employee_id **Foreign Keys:** organization_id, department_id, manager_employee_id

Organization (organization_id (PK), organization_name)

Primary key: Organization_id

Department (Department_id (PK) Department_name)

Primary key: Department_id

Manager (manager_employee_id (PK), manager_employee_name)

Primary key: manager_employee_id,